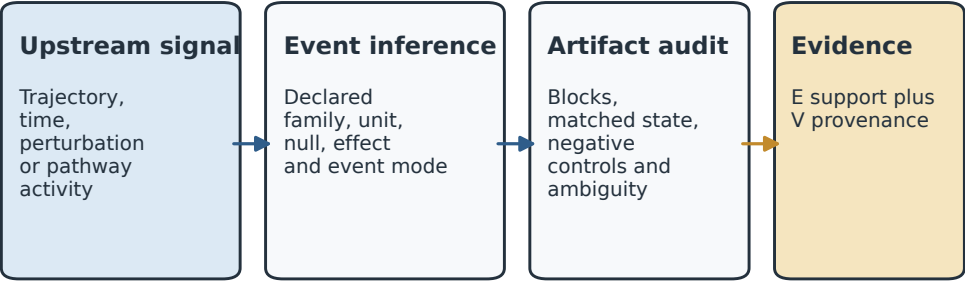


Problem definition and TED algorithm

A. From a signal to an interpretable event



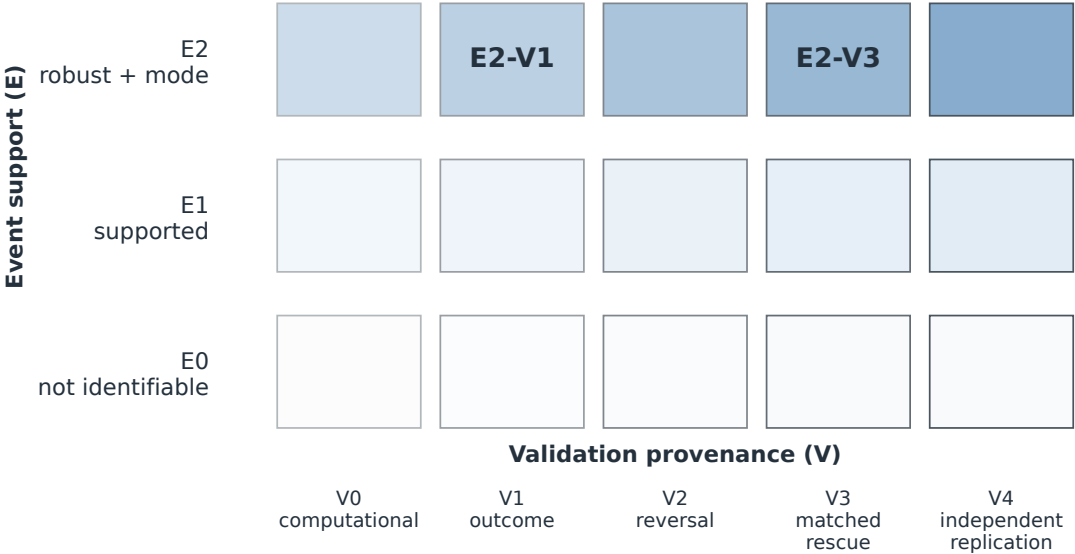
Detection, classification, artifact assessment and validation are separate targets

B. TED event-record fields

Identity	dataset; contrast; event family; analysis unit
Estimate	effect; direction; timing; event-FDR
Robustness	block support; state overlap; control margin
Interpretation	event mode; ambiguity set; identifiability
Evidence	E code; V code; supported and unsupported interpretation

Missing optional evidence is recorded as unavailable, not as negative evidence.

C. Orthogonal evidence descriptor



D. Applicability and fail-closed behavior

Minimum requirements

- Biological blocks
- Declared event family
- Adequate state overlap
- Plausible negative controls
- Interpretable analysis unit

Return E0 / ambiguity

- ✗ No replicate
- ✗ Batch = condition
- ✗ Trajectory not identifiable
- ✗ No defensible null
- ✗ Single smooth curve only

Large cell counts do not replace independent biological units.